FINTECH MDM DATA SCIENCE

HEALTHCARE ANALYTICS DOMAIN

Primary Target: 30-Day Readmission

Chronic Disease Management Analytics: Diabetes

Data Cleaning • Feature Extraction • Outcome Analysis • Input–Output Framework

PROBLEM STATEMENT

"Analysis of chronic disease management trends to identify factors associated with poor disease management and support improved patient monitoring and healthcare outcomes."

PROJECT TEAM

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RESEARCH SCOPE & OBJECTIVES

What Are We Trying to Understand?

Beyond Simple Prediction

Shift focus from diagnosis prediction toward understanding post-discharge outpatient instability and acute relapse patterns.

Core Research Question

Which patient baseline metrics, clinical regimens, and historical visit patterns correlate most strongly with acute care relapses?

Operational Target

30-day hospital readmission serves as the primary quantifiable healthcare proxy for chronic disease management failure.

END-TO-END ANALYTICAL JOURNEY

Patient Baseline
Demographics & Vitals

→

Clinical Regimen
Meds & Diagnoses

→

Care Patterns
Past Utilization

→

30-Day Readmission
Primary Failure Proxy

→

Targeted Protocols
Clinical Action

EVIDENCE BASE SELECTION

Evaluating Our Data Sources

Dataset	Core Strengths	Key Limitations	Project Role
🥇 **UCI Diabetes 130-US**	101,766 encounters, 47 features, real readmission outcomes.	Lacks direct lifestyle habits (diet, step counts).	**PRIMARY DATASET**
🥈 **CDC NHANES**	Rich lifestyle, nutrition, and socioeconomic metrics.	Cross-sectional; no longitudinal readmission tracking.	**Contextual Support**
🥉 **MIMIC-IV**	Granular ICU records with timestamped events.	Excessive multi-table relational complexity.	**Enterprise Reference**
4️⃣ **Kaggle Diabetes**	Pre-packaged adherence and checkup frequency fields.	Synthetic data; cannot serve as real-world evidence.	**Conceptual Reference**

Selection Decision: UCI 130-US is selected as the primary modeling foundation due to its verified hospital readmission outcome labels and multi-facility clinical diversity.

DATA ARCHITECTURE

Primary Dataset Structure & Variable Inventory

101,766
Hospital Encounters

130
US Hospitals

50
Total Columns

47
Clinical Variables

Demographics

12 Variables

patient_nbr, encounter_id, age, gender, race, admission/discharge IDs, time_in_hospital, payer_code, weight.

Diagnostics

9 Variables

diag_1, diag_2, diag_3 (ICD-9 codes), num_lab_procedures, num_procedures, number_diagnoses, A1Cresult, max_glu_serum.

Medications

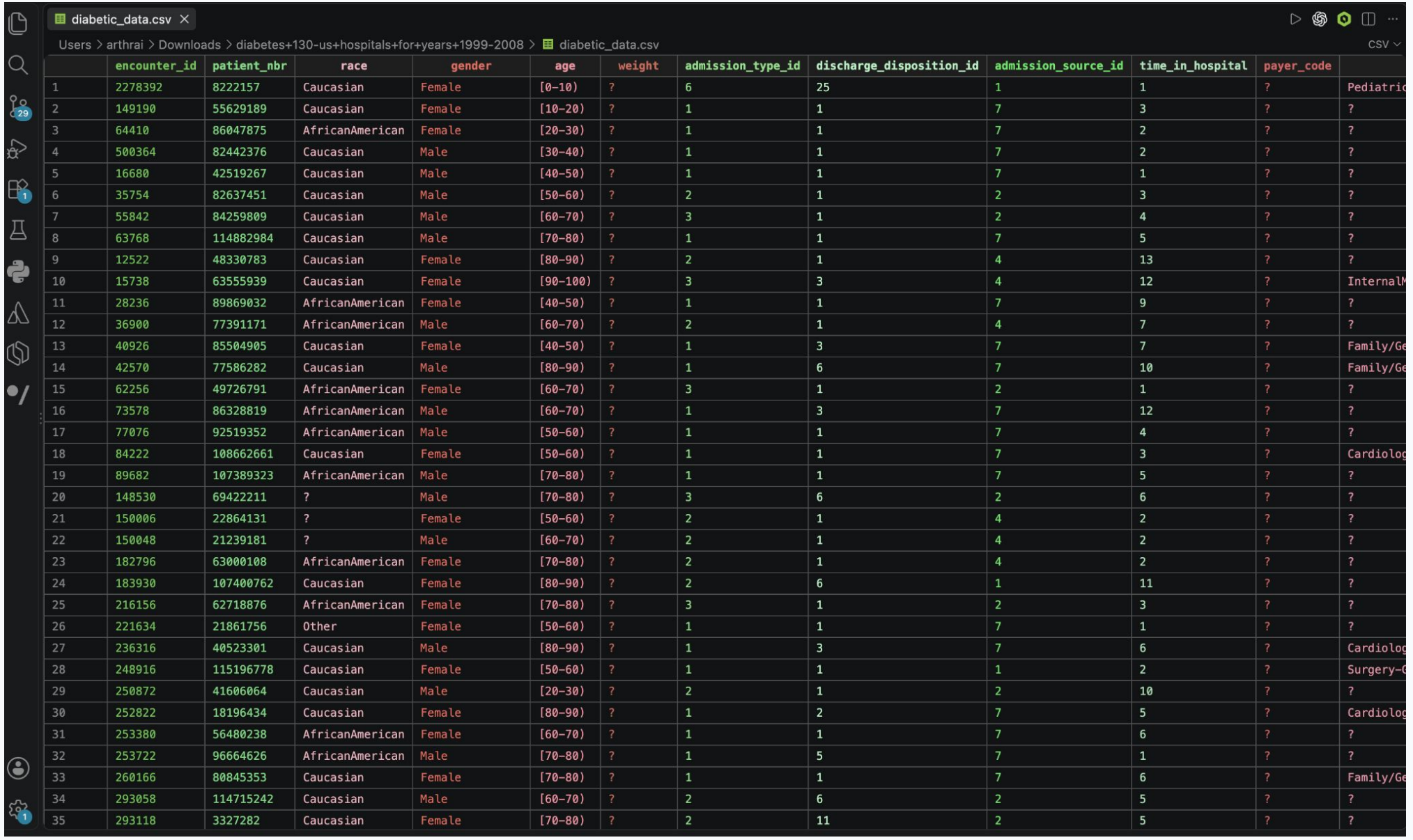
22 Variables

num_medications, 23 specific diabetes drug therapies (insulin, metformin, glipizide), and dosage change flags.

Utilization

4 Variables

Past outpatient, inpatient, and emergency visit counts; 30-day readmission status (NO, >30, <30).


diabetic_data.csv

Users > arthrai > Downloads > diabetes+130-us+hospitals+for+years+1999-2008 > diabetic_data.csv

CSV

	encounter_id	patient_nbr	race	gender	age	weight	admission_type_id	discharge_disposition_id	admission_source_id	time_in_hospital	payer_code	
1	2278392	8222157	Caucasian	Female	[0-10)	?	6	25	1	1	?	Pediatric
2	149190	55629189	Caucasian	Female	[10-20)	?	1	1	7	3	?	?
3	64410	86047875	AfricanAmerican	Female	[20-30)	?	1	1	7	2	?	?
4	500364	82442376	Caucasian	Male	[30-40)	?	1	1	7	2	?	?
5	16680	42519267	Caucasian	Male	[40-50)	?	1	1	7	1	?	?
6	35754	82637451	Caucasian	Male	[50-60)	?	2	1	2	3	?	?
7	55842	84259809	Caucasian	Male	[60-70)	?	3	1	2	4	?	?
8	63768	114882984	Caucasian	Male	[70-80)	?	1	1	7	5	?	?
9	12522	48330783	Caucasian	Female	[80-90)	?	2	1	4	13	?	?
10	15738	63555939	Caucasian	Female	[90-100)	?	3	3	4	12	?	InternalM
11	28236	89869032	AfricanAmerican	Female	[40-50)	?	1	1	7	9	?	?
12	36900	77391171	AfricanAmerican	Male	[60-70)	?	2	1	4	7	?	?
13	40926	85504905	Caucasian	Female	[40-50)	?	1	3	7	7	?	Family/Ge
14	42570	77586282	Caucasian	Male	[80-90)	?	1	6	7	10	?	Family/Ge
15	62256	49726791	AfricanAmerican	Female	[60-70)	?	3	1	2	1	?	?
16	73578	86328819	AfricanAmerican	Male	[60-70)	?	1	3	7	12	?	?
17	77076	92519352	AfricanAmerican	Male	[50-60)	?	1	1	7	4	?	?
18	84222	108662661	Caucasian	Female	[50-60)	?	1	1	7	3	?	Cardiolog
19	89682	107389323	AfricanAmerican	Male	[70-80)	?	1	1	7	5	?	?
20	148530	69422211	?	Male	[70-80)	?	3	6	2	6	?	?
21	150006	22864131	?	Female	[50-60)	?	2	1	4	2	?	?
22	150048	21239181	?	Male	[60-70)	?	2	1	4	2	?	?
23	182796	63000108	AfricanAmerican	Female	[70-80)	?	2	1	4	2	?	?
24	183930	107400762	Caucasian	Female	[80-90)	?	2	6	1	11	?	?
25	216156	62718876	AfricanAmerican	Female	[70-80)	?	3	1	2	3	?	?
26	221634	21861756	Other	Female	[50-60)	?	1	1	7	1	?	?
27	236316	40523301	Caucasian	Male	[80-90)	?	1	3	7	6	?	Cardiolog
28	248916	115196778	Caucasian	Female	[50-60)	?	1	1	1	2	?	Surgery-C
29	250872	41606064	Caucasian	Male	[20-30)	?	2	1	2	10	?	?
30	252822	18196434	Caucasian	Female	[80-90)	?	1	2	7	5	?	Cardiolog
31	253380	56480238	AfricanAmerican	Female	[60-70)	?	1	1	7	6	?	?
32	253722	96664626	AfricanAmerican	Male	[70-80)	?	1	5	7	1	?	?
33	260166	80845353	Caucasian	Female	[70-80)	?	1	1	7	6	?	Family/Ge
34	293058	114715242	Caucasian	Male	[60-70)	?	2	6	2	5	?	?
35	293118	3327282	Caucasian	Female	[70-80)	?	2	11	2	5	?	?

DATA INTEGRITY

Data Cleaning — Missingness & Data Quality Audit

⚠Severe Missingness (Explicit "?" Tokens)

weight — **96.86% Missing**
Dropped completely; prevents extreme imputation bias.

medical_specialty — **49.08% Missing**
Retained as explicit "Missing/General Doctor" category.

payer_code — **39.56% Missing**
Preserved as "Unknown" socioeconomic proxy.

Minor Missingness Imputation

race (2.23%) and diagnosis codes **diag_1 (0.02%), diag_2 (0.35%), diag_3 (1.40%)**: Imputed with modal classes or grouped into "Unknown/Other".

✓100% Complete Clinical Fields (0% Missing)

- **Demographics:** Age, gender
- **Clinical Load:** Medication counts, lab procedures, stay length
- **Utilization History:** Prior outpatient, emergency, inpatient visits
- **Target Variable:** 30-day readmission status complete across all records

OPERATIONAL PROTOCOLS

Data Cleaning Strategy & Operational Rules

Data Quality Issue	Observed Pattern	Treatment Executed
Missing Markers	Missing values stored as "?" string tokens.	**Converted to standard nulls; encoded as explicit "Unknown" categories.**
Hospice / Expired	Discharge codes for expired patients or hospice care (2,423 encounters).	**Excluded from cohort; patients ineligible for readmission (Cohort: 99,343).**
Repeated Patients	Multiple encounter_id entries under a single patient_nbr.	Retained: Chronic disease analytics requires studying cumulative hospital reliance.
Utilization Outliers	Heavy right skew in acute care visits (number_emergency > 30).	Retained: Captures authentic high-risk behavior rather than noise.
Type Encoding	Administrative IDs stored as integers (admission_type_id).	Converted to explicit categorical factors to prevent artificial numeric ordinality.

Analytics Principle: Patient exclusions are restricted strictly to non-eligible discharge codes (hospice/expired) to preserve clinical validity while maintaining authentic real-world outlier distributions.

ANALYTICAL ENGINEERING

Feature Extraction Pipeline

1. Binary 30-Day Readmission (readmitted_30d)

```
readmitted_30d = 1 if readmitted == "<30" else 0
```

Establishes the primary acute care relapse target (11,314 positive cases, **11.39% baseline rate**).

2. Prior Healthcare Utilization Index

```
total_previous_visits = number_outpatient + number_inpatient + number_emergency
```

Aggregates past acute reliance into one composite metric of patient healthcare system dependence.

3. Polypharmacy Index (Binned Complexity)

Low (0–5 meds)	Moderate (6–14)	High (15+ meds)
4,938 enc (7.61%)	43,526 enc (10.28%)	50,879 enc (12.70%)

4. ICD-9 Clusters & Titration Indicator

- **ICD-9 Mapping:** High-cardinality diag_1 mapped to 4 clusters: Circulatory/Heart, Direct Diabetes, Respiratory, Other.
- **medication_changed:** Binary flag isolating whether any diabetes drug dosage was altered during inpatient stay.

THEORETICAL MODEL

The Core Framework: Input → Channel → Output

STAGE 1 **Observable Inputs**

Patient & Clinical Baseline

- **Demographics:** Age bracket, gender, race
- **Clinical Status:** Diagnoses clusters, lab procedures, stay length
- **Medication Regimen:** Drug count, insulin therapy, dose adjustments
- **Prior Utilization:** Past emergency, outpatient, and inpatient stays



STAGE 2 **Analytical Channel**

Connecting Mechanism

Disease Management & Care Utilization Patterns

Evaluates how well the condition is stabilized in outpatient ambulatory settings versus acute emergency/inpatient over-reliance.



STAGE 3 **Healthcare Output**

Measurable Endpoint

30-Day Readmission

$Y \in \{0, 1\}$

Indicates post-discharge chronic management failure and acute clinical relapse.

Goal: Translate observable input anomalies into early warning signals that identify failure within the care channel before acute readmission occurs.

HYPOTHESIS FORMULATION

Candidate Explanatory Variables

Dimension	Variables Examined	Research Question
Demographic	age, gender, race	Do readmission rates differ significantly across age brackets and demographic cohorts?
Clinical	diag_1 clusters, number_diagnoses, time_in_hospital, A1Cresult	Does comorbidity burden or stay duration correlate with higher readmission risk?
Treatment	num_medications, insulin, medication_changed	Does polypharmacy or active medication alteration reflect clinical instability?
Utilization	number_inpatient, number_emergency, total_previous_visits	Does frequent prior hospital utilization act as a leading indicator of relapse?

Methodological Stance: Hypotheses evaluate statistical associations and risk stratification power while explicitly controlling for confounding variables in multivariable regression.

VERIFICATION PIPELINE

Statistical Testing Roadmap

STEP 01

Descriptive Baseline

Summarize distribution metrics (mean, median, IQR) and confirm 11.39% readmission cohort baseline.

STEP 02

Bivariate Stratification

Contrast readmission rates directly across candidate factors (visit counts, polypharmacy tiers).

STEP 03

Hypothesis Testing

Pearson's Chi-Square (χ^2) for categorical traits; non-parametric Mann–Whitney U for skewed counts.

STEP 04

Multivariable Logit

Fit adjusted models to compute Odds Ratios (aOR) with 95% CIs while controlling for clinical confounders.

STEP 05

Clinical Translation

Convert statistically significant odds ratios into actionable risk tiers and discharge protocols.

Significance Threshold: All tests evaluated at significance level **$\alpha = 0.05$**. Non-parametric methods applied to avoid bias from heavily right-skewed acute care utilization distributions.

EMPIRICAL EVIDENCE

Exploratory Visuals & Key Observations

Prior Hospital Stays

22.02% vs 8.59% (0 stays)

Patients with 2+ prior inpatient stays face over 2.5× higher readmission rate ($p < 0.001$).

Takeaway: Strongest single relapse indicator.

Polypharmacy Load

12.70% High (15+ meds)

High polypharmacy exhibits 12.70% rate vs 7.61% for low-regimen patients ($\chi^2 = 209.17$, $p < 0.001$).

Takeaway: Reflects frailty & dosing vulnerability.

Age & Stay Length

12.47% Age 80+ bracket

Geriatric patients show elevated readmission ($\chi^2 = 133.60$, $p < 0.001$) with stay lengths of 3–6 days.

Takeaway: Burden concentrates in multimorbid groups.

HbA1c & Titration

$p < 0.001$ $\chi^2 = 42.56$

Significant association between A1C result and readmission outcomes across cohorts.

Takeaway: Active testing mitigates post-discharge risk.

All exploratory associations confirmed statistically significant at $p < 0.001$, motivating rigorous multivariable modeling to eliminate confounding effects.

STATISTICAL VALIDATION & MULTIVARIABLE MODELING

Empirical Findings & Multivariable Logistic Regression

Cohort: 99,343 Encounters

Readmissions: 11,314 (11.39%)

HYPOTHESIS TESTS (ALL P < 0.001): **Inpatient: MW U = 6.05×10^8** **ER Visits: MW U = 5.30×10^8** **Polypharmacy: $\chi^2 = 209.17$** **Insulin Status: $\chi^2 = 199.30$** **Age Groups: $\chi^2 = 133.60$**

Predictor Variable	Adjusted OR	95% CI	p-value	Clinical Interpretation
Prior Inpatient Visits	**1.310**	1.293–1.327	**<0.001**	**+31.0% risk** per prior admission; primary relapse predictor.
Prior Emergency Visits	**1.034**	1.017–1.052	**<0.001**	Each prior ER visit increases adjusted odds by 3.4%.
Diagnosis Count	**1.048**	1.035–1.060	**<0.001**	Comorbidity burden adds ~4.8% incremental risk per condition.
Length of Stay	**1.024**	1.016–1.031	**<0.001**	Extended stay duration reflects acute clinical frailty.
Age 80+ (vs. 0–39)	**1.219**	1.106–1.345	**<0.001**	Geriatric patients face 21.9% higher adjusted relapse odds.
Age 60–79 (vs. 0–39)	**1.138**	1.039–1.247	**0.006**	Older adult diabetic cohorts face 13.8% higher adjusted odds.
Medication Changed	**0.936**	0.887–0.988	**0.016**	**6.4% protective effect** from active inpatient titration.

Multivariable Logistic Model (Logit Link) | McFadden Pseudo-R^2 = 0.0363 | Dependent Variable: readmitted_30d

ACTIONABLE OUTCOMES

Translating Analytics into Healthcare Interventions

PRIORITY 1

Frequent Hospital Users

EMPIRICAL SIGNAL

Prior inpatient aOR = 1.310 (+31% / visit). Readmission reaches **22.02%** for ≥2 visits.

INTERVENTION PROTOCOL

Assign dedicated case managers; mandate automated 48-hour post-discharge phone follow-ups and rapid outpatient check-ins.

PRIORITY 2

Complex Polypharmacy

EMPIRICAL SIGNAL

High polypharmacy (15+ meds) drives readmission to **12.70%** (χ^2 = 209.17).

INTERVENTION PROTOCOL

Mandate discharge medication reconciliation with a clinical pharmacist to resolve adverse drug interactions and dosing errors.

PRIORITY 3

Geriatric Comorbidity

EMPIRICAL SIGNAL

Age 80+ aOR = 1.219; diagnosis count aOR = 1.048 per co-occurring chronic condition.

INTERVENTION PROTOCOL

Co-schedule multidisciplinary cardiology, nephrology, and endocrine outpatient reviews within 7 days of discharge.

Strategic Shift: Moves health system operations from reactive emergency readmission care to proactive, algorithmic chronic disease management.

MILESTONE REVIEW

Summary & Analytical Next Steps

Completed Milestones

Data Ingestion & Cohort Filter: Processed 101,766 raw records; removed 2,423 hospice/expired encounters to isolate an analytical cohort of 99,343.

Feature Engineering: Constructed derived metrics (readmitted_30d, total_previous_visits, polypharmacy bins, ICD-9 clusters).

Statistical Testing: Confirmed significant associations across age, polypharmacy, and utilization via Chi-Square and Mann–Whitney U tests (all $p < 0.001$).

Multivariable Logit: Fitted logistic regression isolating prior inpatient visits (aOR = 1.310) as the leading independent risk factor.

Immediate Next Steps

Risk Scoring Matrix: Finalize a rule-based patient risk scoring matrix (Tier 1 High, Tier 2 Moderate, Tier 3 Low) based on empirical logit coefficients.

Decision-Support Workflows: Construct workflow mockups illustrating pharmacist reconciliation alerts and 48-hour outpatient outreach triggers.

Model Packaging & Calibration: Package final code, model calibration plots, ROC curves, and clinical translation recommendations for final delivery.